

Kata Containers: Secure Containers from the OpenStack Community

Kata Containers is new container technology that combines technology from Intel Clear Containers with runV from Hyper. Managed by a community, the objective of Kata is to deliver speed and security.



Ever since Linux containers were launched with Docker, containerisation has become a full-fledged domain. Containers are now used in a number of applications and, as in the case of all developing technologies, real-life challenges linked to performance and security have begun to matter with containers as well.

Intel has been working on the Clear Containers Project for some time to address security concerns within containers through Intel Virtualization Technology (Intel VT). This essentially offers the capability to launch containers as lightweight virtual machines (VMs), providing an alternative runtime, which is interoperable with popular container environments such as Kubernetes and Docker. At the same time, the Hyper community has been working on providing the alternate OCI-compliant runtime to run containers on hypervisors, with a few limitations caused by the current incompatibility between hypervisors and containers.

In recent times, it's been noticed that single vendor open source projects and communities do not attract many contributors due to their inherent vendor-specific policies. There has always been a need for an open source community to build these projects up, along with the current set of contributors, and drive further collaborative innovation.

During the last KubeCon, OpenStack Foundation announced a new initiative aimed at unifying the speed and manageability of containers with the security advantages of virtual machines (VMs). This was called Kata Containers.

What are Kata Containers?

Kata Containers is an open source project and community, working to build a standard implementation of lightweight virtual machines (VMs) that feel and perform like containers, but provide the workload isolation and security advantages of VMs. Intel is contributing Intel Clear